

Arkaprava Sinha | Ph.D. Candidate

University of North Carolina, Charlotte

+1(704)-352-3502 • asinha13@charlotte.edu • thearkaprava.github.io

Research Summary

I am a Ph.D. candidate at UNC Charlotte, where I work on getting machines to understand what happens in long, real-world video. Most of my research sits at the intersection of temporal modeling, multimodal learning, and agentic reasoning. I build architectures that can process extended, untrimmed footage from both egocentric and third-person viewpoints without choking on the sequence length. This line of work has led to publications at CVPR (2025, 2026) and AAAI (2025), with ongoing projects on agentic temporal reasoning and unified egocentric representations. I also serve as a consultant researcher with the Lowe's Computer Vision Team at UNCC. Previously, I spent 2+ years in industry shipping CV/ML systems for Fortune 50 clients.

Research Interests: Multimodal Learning · Agentic AI · Long Video Understanding · Egocentric Video · State-Space Models · Vision-Language Models · Temporal Representation Learning

Education

University of North Carolina, Charlotte Ph.D. Computer Science (Advisor: Dr. Srijan Das)	2023–Present
Chennai Mathematical Institute M.Sc. Data Science	2018–2020
University of Calcutta B.Sc. (Hons.) Statistics	2015–2018

Experience

University of North Carolina, Charlotte Graduate Research Assistant – Charlotte Vision Lab	Aug 2023 – Present
- Agentic temporal reasoning framework that iteratively proposes and verifies temporal hypotheses, enabling efficient long-video question answering without exhaustive frame sampling. [1] - Unified egocentric video representation learning method leveraging proxy tasks as mediators for self-supervised pretraining on ego-centric data. [2] - Multi-scale State-Space Model (Mamba) architecture for long untrimmed video understanding, establishing new state-of-the-art on temporal action detection and video summarization benchmarks. [3] - Multimodal (RGB + Human Pose + Human Object Interactions) Large Language Model for understanding daily living actions from video, with curriculum-based training for integrating new modalities. [5] - Multimodal (RGB + Skeleton + Language) Vision-Language Model that integrates skeleton motion into a joint embedding space for robust activity recognition. [6] - Deployed a real-time activity detection model in retail stores of a Fortune 50 company for optimized resource allocation.	
Accenture AI Data Scientist	Nov 2021 – Feb 2023
- Applied advanced NLP and transformer-based models to analyze large-scale financial documents, reducing manual review effort and accelerating insight generation for enterprise clients. - Developed and deployed scalable ML algorithms for Accenture's AI solutions library, strengthening reusable assets and speeding up delivery for downstream projects.	
LTIMindtree (Formerly Larsen & Toubro Infotech) Data Scientist	Jul 2020 – Nov 2021
- Designed industry-scale Computer Vision solutions for Fortune 50 clients—including vision-enabled security systems and real-time anomaly detection pipelines. - Built Deep Learning models for the Automated Modeling Platform, increasing model reliability and accelerating end-to-end deployment.	
Teradata India Pvt. Ltd. Data Science Intern	May 2019 – Jul 2019
Developed topic modeling frameworks for Teradata's ML engine and researched advanced clickstream analysis approaches.	

Publications

(C)onference (P)reprint (W)orkshop

1. (P) **Arkaprava Sinha**, Dominick Reilly, Siddharth Krishnan, Hieu Le and Srijan Das, "TIMEPROVE: Propose, then Verify for Efficient Long Video Temporal Reasoning in Activities of Daily Living," arXiv, 2026.
2. (P) Wenhao Chi*, **Arkaprava Sinha***, Dominick Reilly, Hieu Le and Srijan Das, "UNIEGO: Proxies as Mediators for Unified Egocentric Video Representation Learning," arXiv, 2026.
3. (C) **Arkaprava Sinha**, Monish Soundar Raj, Ahmed Helmy, Pu Wang, Hieu Le and Srijan Das, "MS-Temba: Multi-Scale Temporal Mamba for Understanding Long Untrimmed Videos," IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
4. (P) Weston Bondurant, **Arkaprava Sinha**, Hieu Le, Srijan Das, Stephanie Schuckers, "DiffSwap++: 3D Latent-Controlled Diffusion for Identity-Preserving Face Swapping," arXiv, 2026.
5. (C) Dominick Reilly, Rajat Subhra Chakraborty, **Arkaprava Sinha**, Manish Kumar Govind, Pu Wang, Francois Bremond, Le Xue and Srijan Das, "LLAVIDAL: A Large Language Vision Model for Daily Activities of Living," IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
6. (C) **Arkaprava Sinha**, Dominick Reilly, Pu Wang, Francois Bremond, and Srijan Das, "SKI Models: Skeleton Induced Vision-Language Embeddings for Understanding Activities of Daily Living," Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2025.
7. (W) Mahmoud Ali, Di Yang, **Arkaprava Sinha**, Dominick Reilly, Srijan Das, Gianpiero Francesca, and Francois Bremond, "Quo Vadis, Video Understanding with Vision-Language Foundation Models?," Workshop on Video-Language Models (NeurIPS), 2024.

Key Research Projects

Agentic Temporal Reasoning for Video

- An **agentic propose-then-verify framework** that decomposes complex temporal queries into sub-goals, iteratively navigating long videos to localize answers.
- Efficient temporal grounding and QA over hour-long videos without exhaustive processing.

Long Video Understanding with State-Space Models

- Efficient spatio-temporal representations using **Mamba/SSM architectures** that scale linearly with sequence length for processing extended untrimmed videos.
- Multi-scale temporal modeling capturing short-term atomic actions and long-range dependencies for temporal action detection and video summarization.

Unified Egocentric Video Representation Learning

- Self-supervised pretraining for **egocentric video** using proxy-mediated contrastive learning across hand-object interactions and action contexts.

Multimodal Vision-Language Models for Activity Understanding

- **Multimodal fusion architectures** integrating RGB, skeleton motion, and language for robust activity recognition across diverse viewpoints.

Technical Skills

- **Core Frameworks:** PyTorch, HuggingFace Transformers, DeepSpeed, FSDP, PyTorch Lightning
- **Research Areas:** State-Space Models (Mamba), Vision-Language Models, Multimodal Fusion, Video Transformers, Self-Supervised Learning
- **Tools & Infrastructure:** CUDA, Slurm, Weights & Biases, Docker, Git, Linux
- **Languages:** Python, C++, R · **Libraries:** OpenCV, Scikit-learn, TensorFlow, FFmpeg